How would an economist solve climate change? (Part II)

ECON-4210

Economics of Climate Change

Class 17

Prof. Casey Wichman

Georgia Institute of Technology

Today's class

1. Questions?
2. Attendance
3. Housekeeping
    1. In-class policy "game" on Monday
    2. Problem set 3 will be posted next week (due following Friday)
4. How would an economist solve climate change?
    1. Comparing price (carbon taxes) vs. quantity (e.g., cap-and-trade) regulations
5. For next class…

Economists tend to favor market-based solutions…

- "Getting the prices right" sends clear signals to polluting firms.
  - Those with the lowest abatement costs will reduce pollution "first"
- This same signal incentivizes firms to reduce pollution as cheaply and generates technological innovation.
- Carbon taxes generate revenue, cap-and-trade does not (although permits can be auctioned off, initially).

Prices vs. Quantities (the "Weitzman rule")

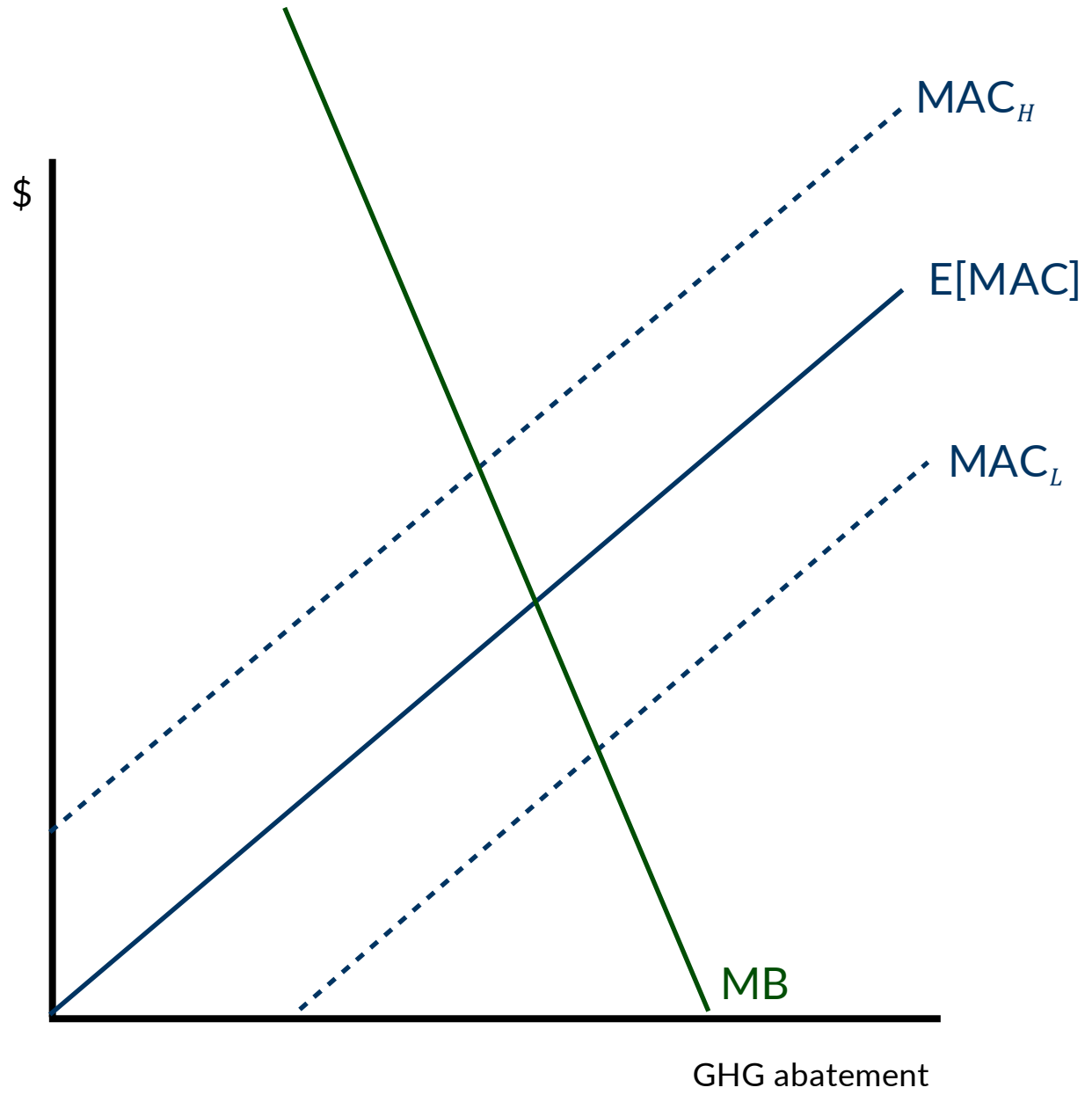
- Economists tend to prefer prices to regulate externalities
- Regulators tend to prefer command-and-control-style regulations

- There is no *theoretical* reason to prefer taxes over quantities
  - Taxes = Pigouvian taxes
  - Quantities = cap-and-trade or command-and-control

- **Under what circumstances should we prefer price vs. quantity regulation?**
  - …it depends on the relative slopes of the marginal abatement cost curves and marginal benefit curves.
  - With no uncertainty, price and quantity controls are equivalent.
  - Only uncertainty about marginal abatement costs matters
    - Regulators typically do not know marginal abatement costs

Prices vs. quantities, graphically…

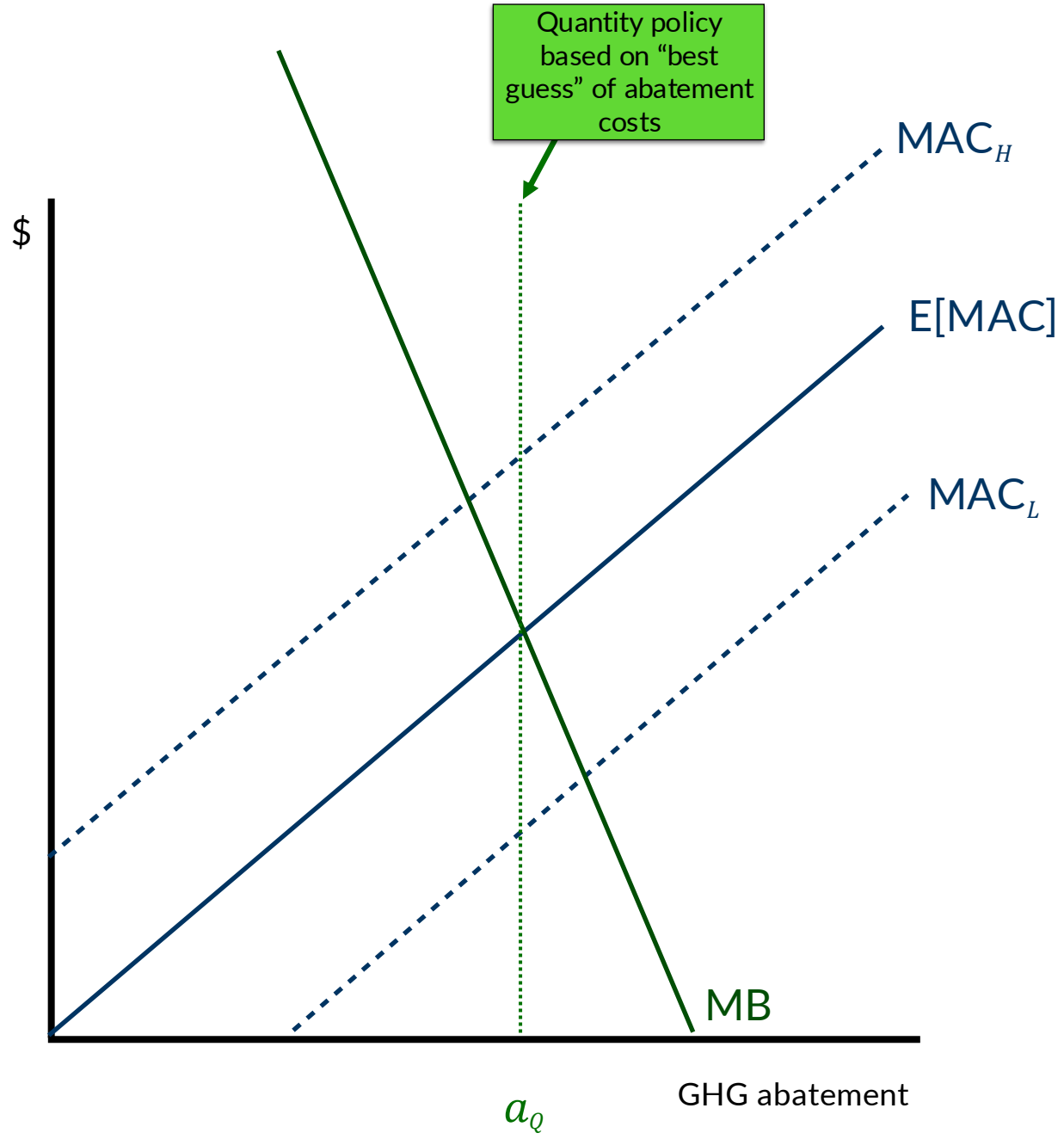
- **<u>Case 1a:</u> Quantity policy with a steep MB curve.**
- MACs are unknown to the regulator.
- Expected MAC (E[MAC]) is the regulator's best guess, but it could be higher (MAC_H) or lower (MAC_L).



Prices vs. quantities, graphically…

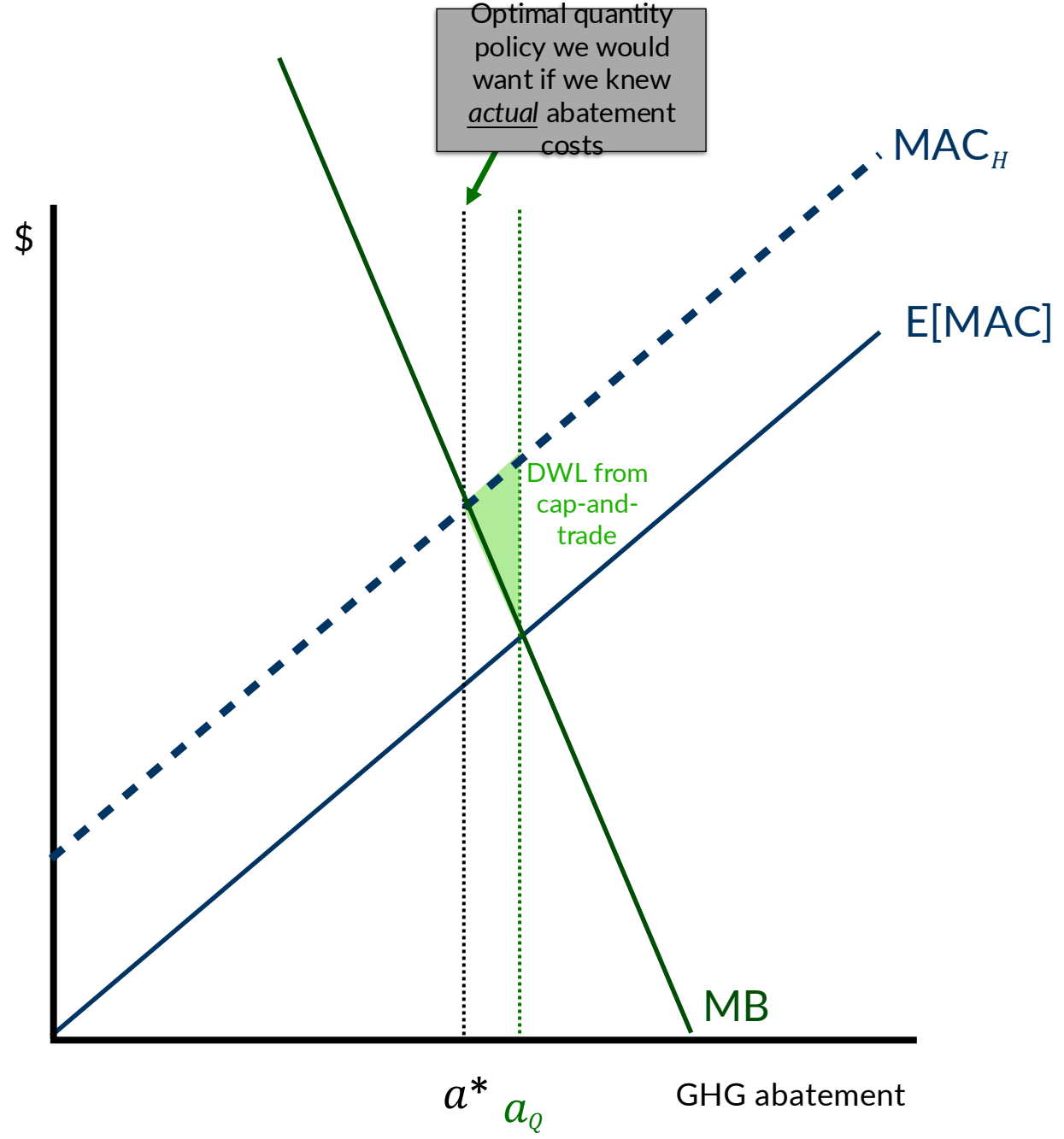
- **Case 1a: Quantity policy with a steep MB curve.**
- MACs are unknown to the regulator.
- Expected MAC (E[MAC]) is the regulator's best guess, but it could be higher (MAC_H) or lower (MAC_L).

- The regulator needs to set a quantity policy (e.g., the "cap" in a cap-and-trade policy) based on their "best guess" of marginal abatement costs.
  - Choose a_Q based on where *expected* MAC intersects with MB curve.
  - This could be a suboptimal level of abatement.



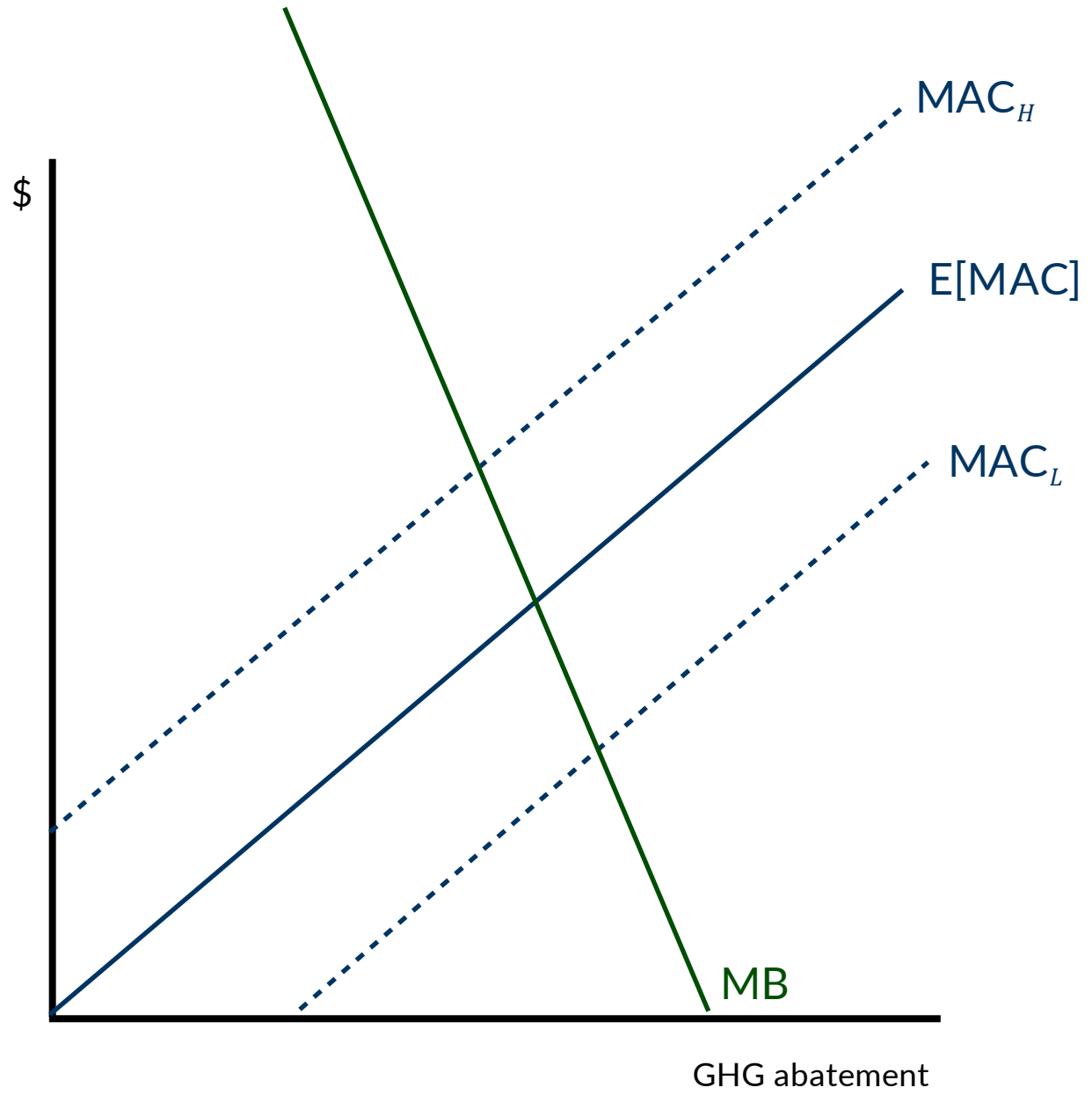
Prices vs. quantities, graphically…

- **<u>Case 1a:</u> Quantity policy with a steep MB curve.**
- If we actually experience MAC_H
  - $\Longrightarrow$ optimal level of abatement is actually a^*
  - *We should have set the cap at a^* where MAC_H = MC*
- So, we have $a_Q > a^*$
  - We have too much abatement under quantity policy at a_Q
- This results in deadweight loss!
  - We abated too much pollution, which imposes extra costs on society.
  - At a_Q, MAC_H > MB (i.e., the marginal costs of abatement exceed the marginal benefits)



Prices vs. quantities, graphically…

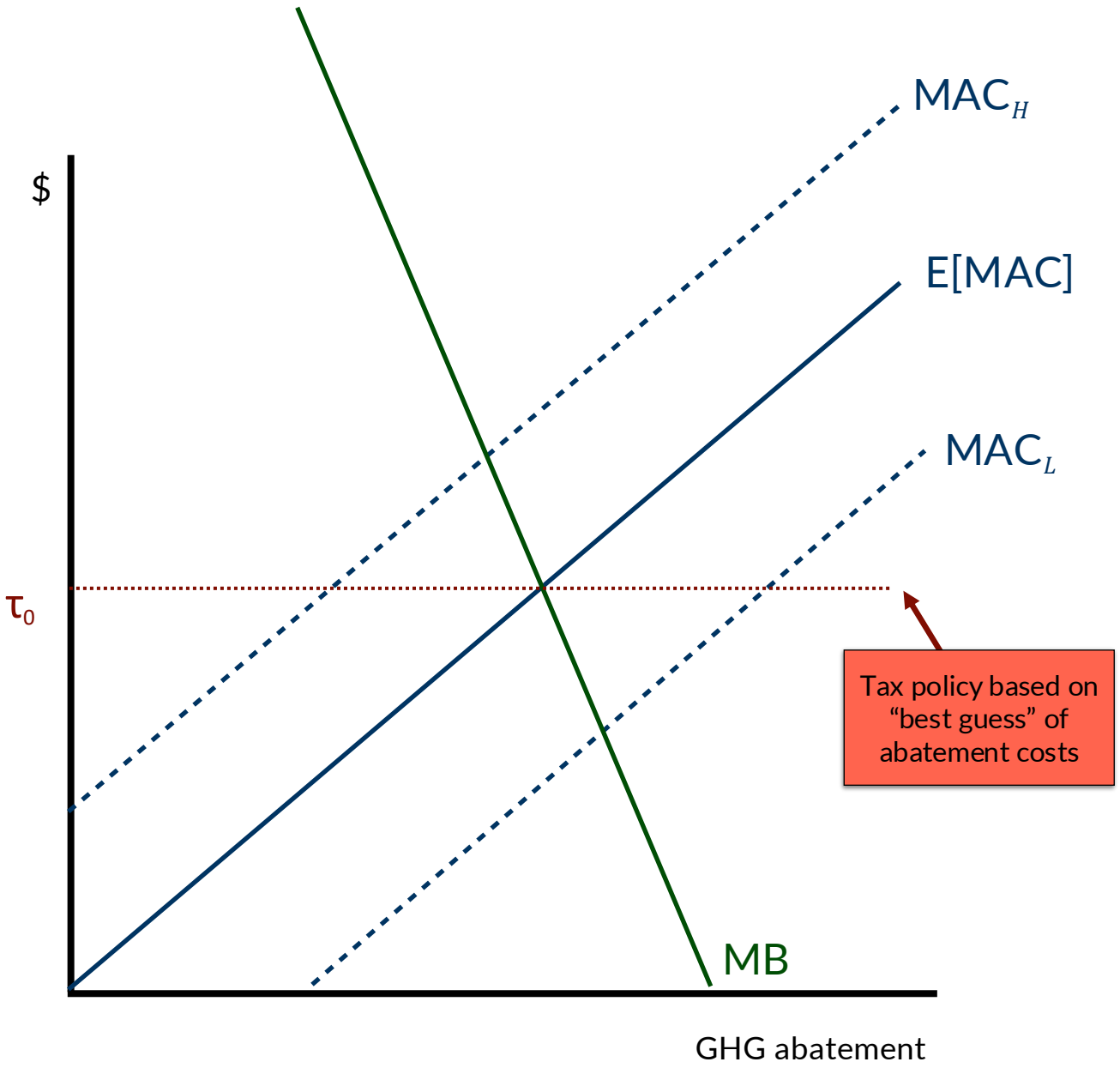
- **<u>Case 1b:</u> Price policy with a steep MB curve.**
- MACs are unknown to the regulator.
- Expected MAC (E[MAC]) is the regulator's best guess, but it could be higher (MAC_H) or lower (MAC_L).



Prices vs. quantities, graphically…

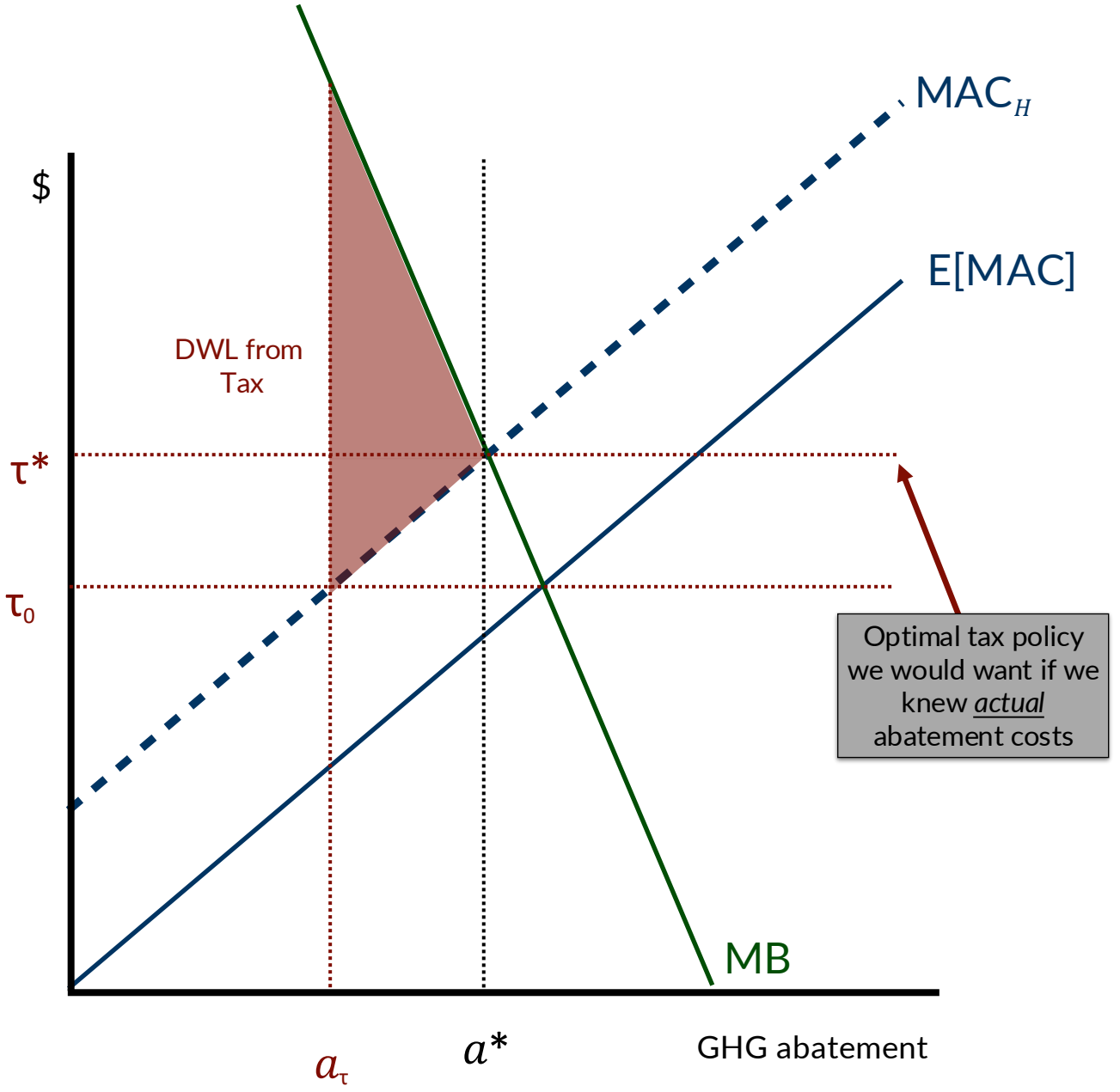
- **Case 1b: Price policy with a steep MB curve.**
- MACs are unknown to the regulator.
- Expected MAC (E[MAC]) is the regulator's best guess, but it could be higher (MAC_H) or lower (MAC_L).

- The regulator needs to set a price policy (e.g., a carbon tax) based on their "best guess" of marginal abatement costs.
  - Choose τ_0 based on where *expected* MAC intersects with MB curve.
  - This could be a suboptimal level of abatement.



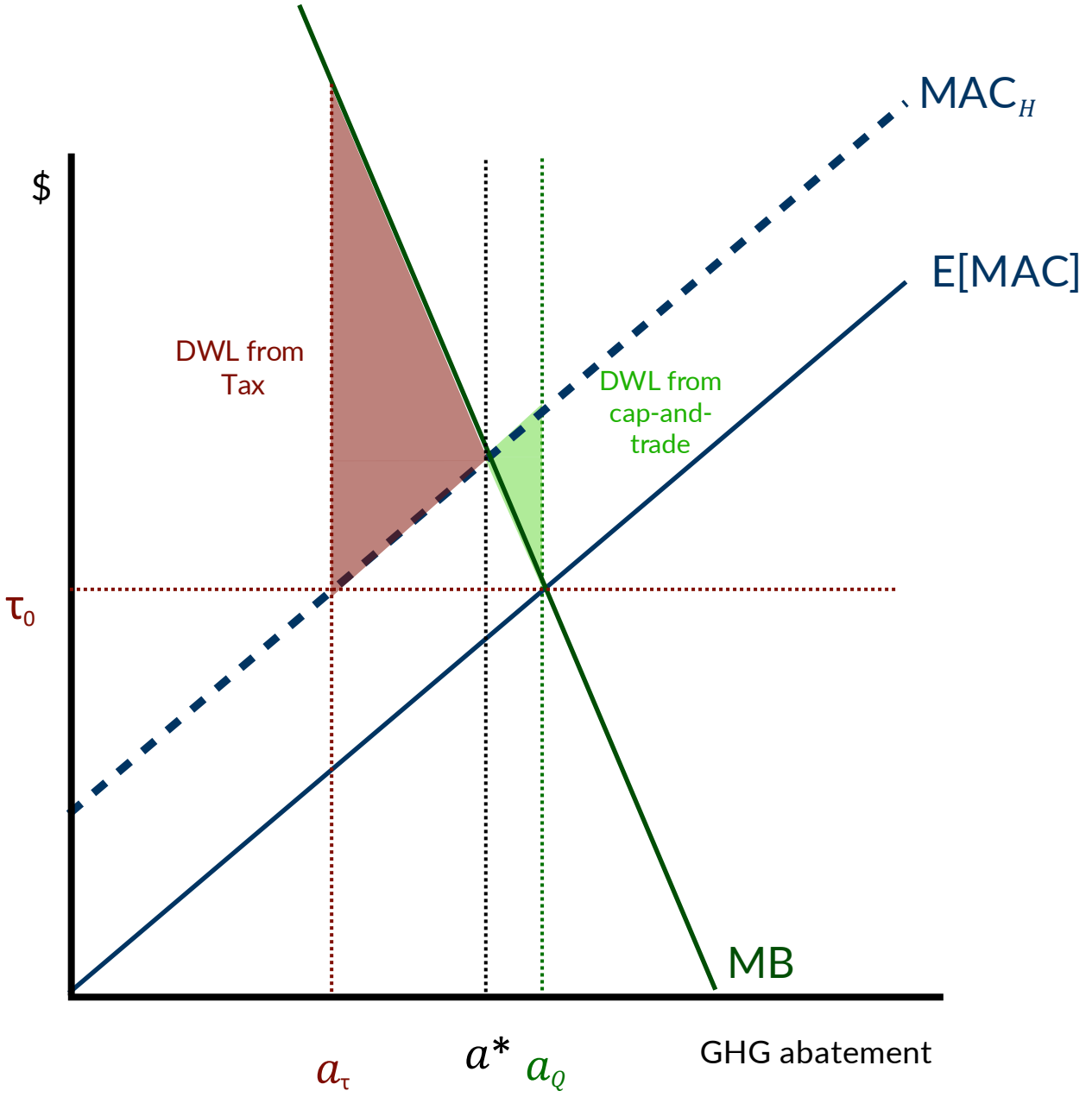
Prices vs. quantities, graphically…

- **Case 1b: Price policy with a steep MB curve.**
- If we actually experience MAC_H
  - $\Longrightarrow$ optimal level of abatement is actually a^*
  - *We should have set tax at* τ^*.
- So, $a_\tau < a^*$
  - We have too little abatement under tax policy at τ_0
- This results in deadweight loss!
  - We abated too little pollution, which imposes extra costs on society.
  - We were willing to pay more for more abatement.
  - At a_τ, MB > MAC_H (i.e., the marginal benefits of abatement exceed the marginal costs)



Prices vs. quantities, graphically…

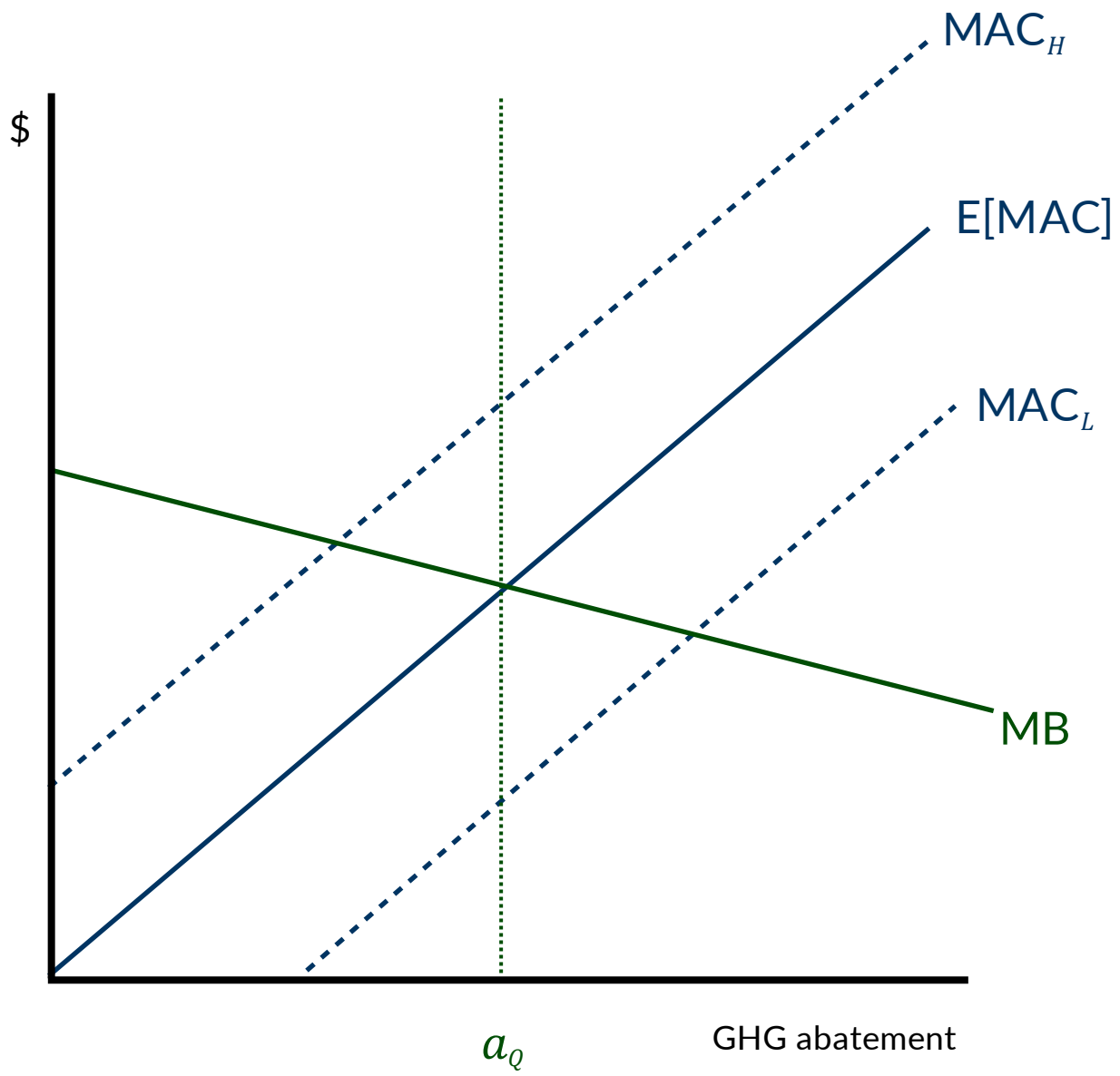
- **Comparing price and quantity controls with steep MB curve.**
- When MAC is higher than expected
  - Tax policy generates too little abatement ($a_\tau < a^*$).
  - Quantity policy generates too much abatement ($a^* < a_Q$).
- With steep MB curve, DWL from tax is much larger than DWL from cap-and-trade.
  - $\Longrightarrow$ **With uncertain MACs, we should prefer cap-and-trade to taxes when MB curve is relatively steep (compared to MAC curve).**



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Prices vs. quantities, graphically…

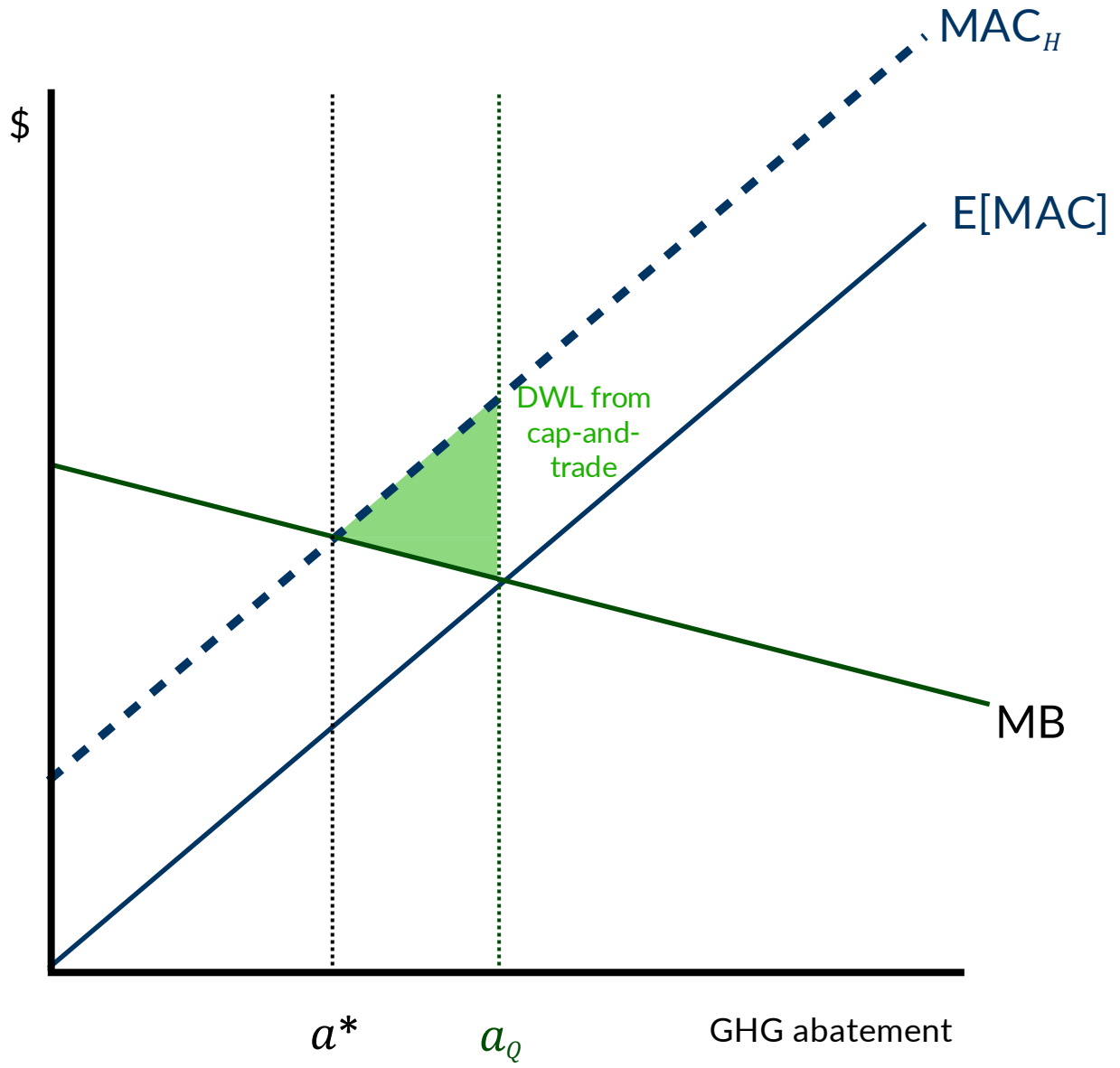
- Case 2: Repeat this exercise with a relatively flat MB curve.



Prices vs. quantities, graphically…

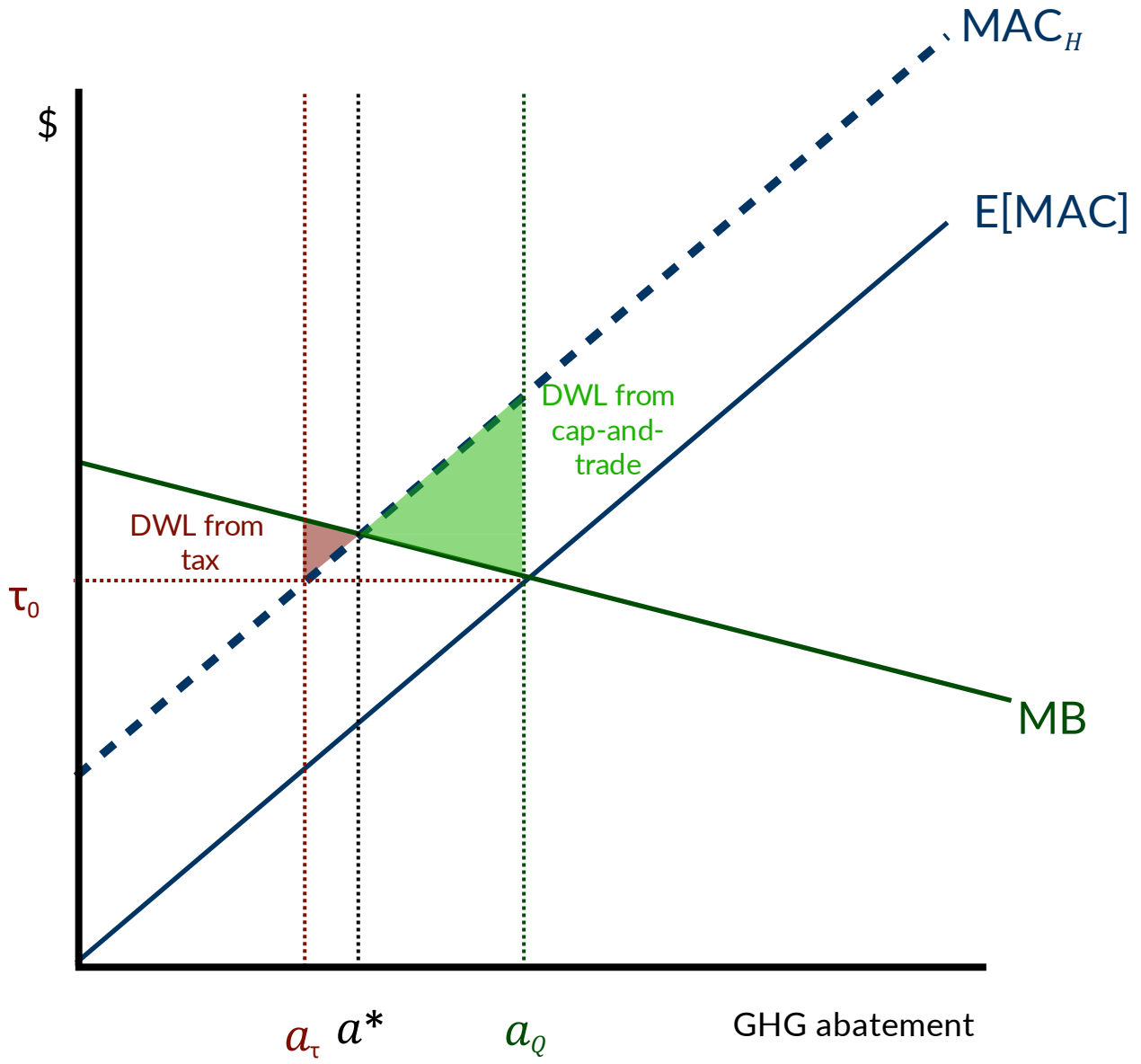
- **Case 2: Repeat this exercise with a relatively flat MB curve.**

- Quantity regulation results in green DWL triangle.
  - Again, too much abatement.
  - MAC_H > MB at a_Q.



Prices vs. quantities, graphically…

- **Case 2: Repeat this exercise with a relatively flat MB curve.**
- Quantity regulation results in green DWL triangle.
  - Again, too much abatement.
  - MAC_H > MB at a_Q.
- Tax results in red DWL.
  - Too little abatement.
  - MB > MAC_H at a_τ.
- **$\Longrightarrow$ with uncertain MACs, tax controls are preferred to quantity regulations when MB is relatively flat.**

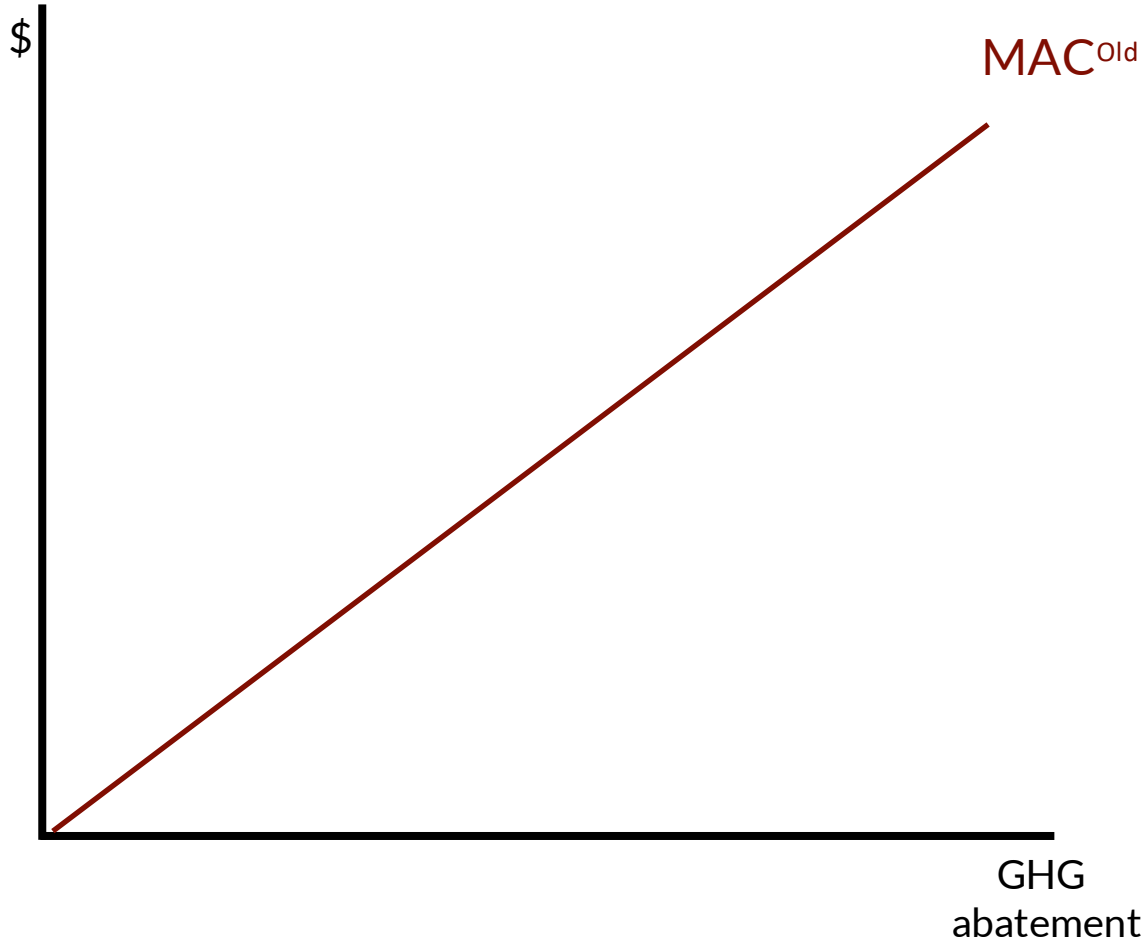


Key takeaways

- When costs are uncertain…
  - If we have relatively steep MB (e.g., slightly more emissions generate large decreases in benefits), then quantity controls are preferred.
    - This makes sense… we can better control the max quantity by setting a cap.
  - If we have relatively flat MB, then price controls are preferred (e.g., like a carbon tax).
- Climate change tends to fall into the "flatter" MB world, where taxes are preferred.
  - Reducing emissions a little bit does not dramatically increase benefits.
  - But, there are other factors that determine policy, e.g., political feasibility.
  - Throughout the world, there is a mix of carbon prices (e.g., British Columbia) and cap-and-trade programs for GHGs (e.g., EU Emissions Trading Scheme).

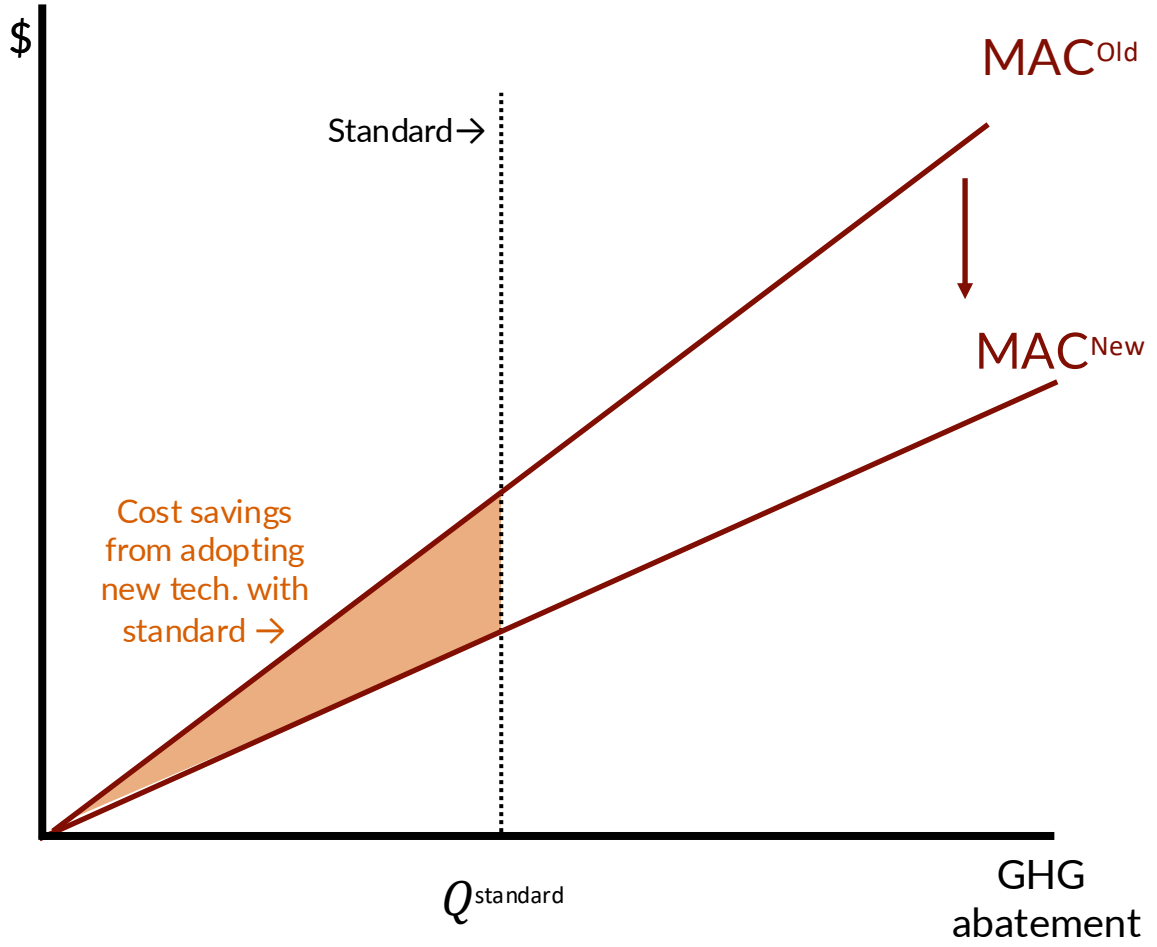
Innovation

- One reason economists prefer market-based policies (and taxes, specifically) is due to innovation.
- Innovation, here, refers to any technology that reduces abatement costs (e.g., generating same output at lower costs).



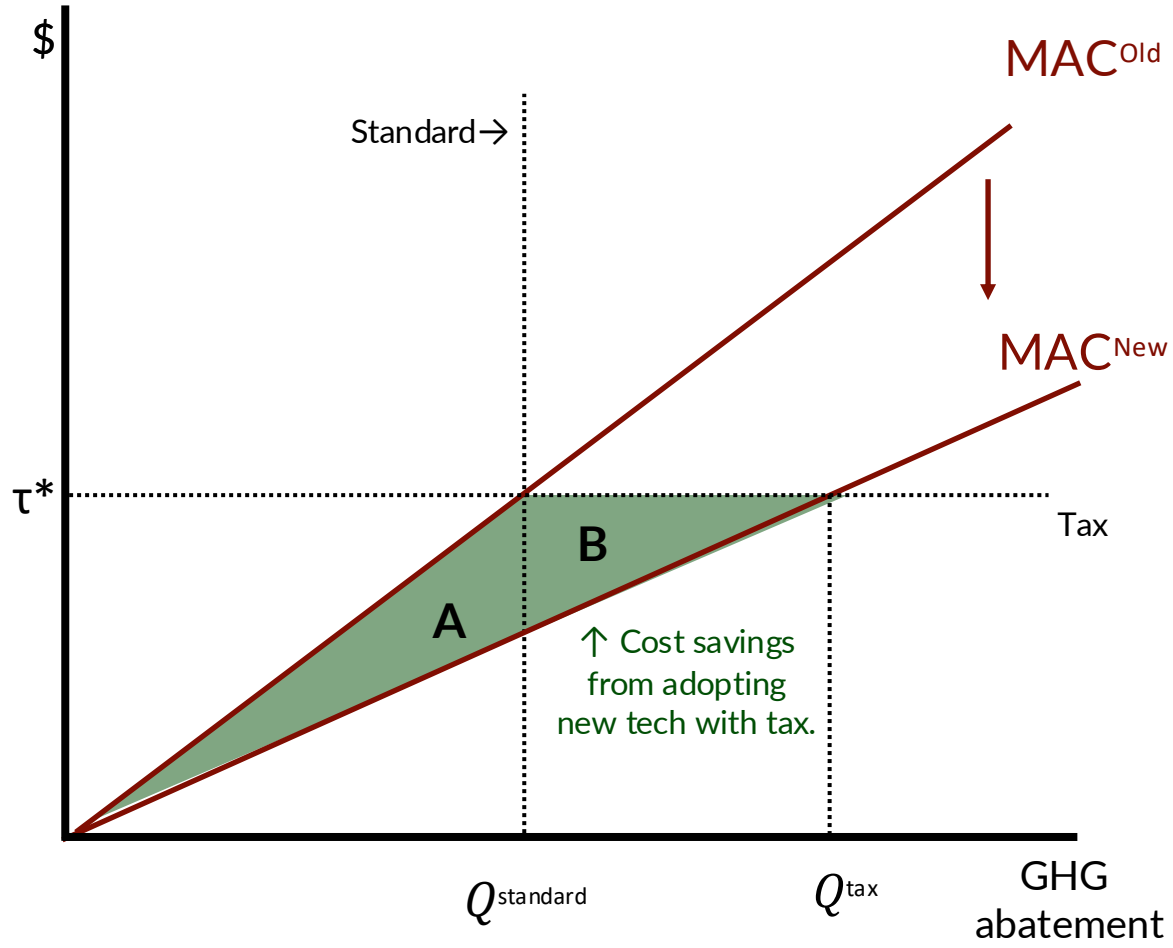
Innovation with standards

- Imagine new technology becomes available that allows firms to generate same output with fewer CO_2 emissions.
- MAC curve pivots downwards.
- Under a standard:
  - Gain to the firm from adopting new technology is given by cost-savings represented by difference between MAC curves.
  - No incentive to abate more than required.



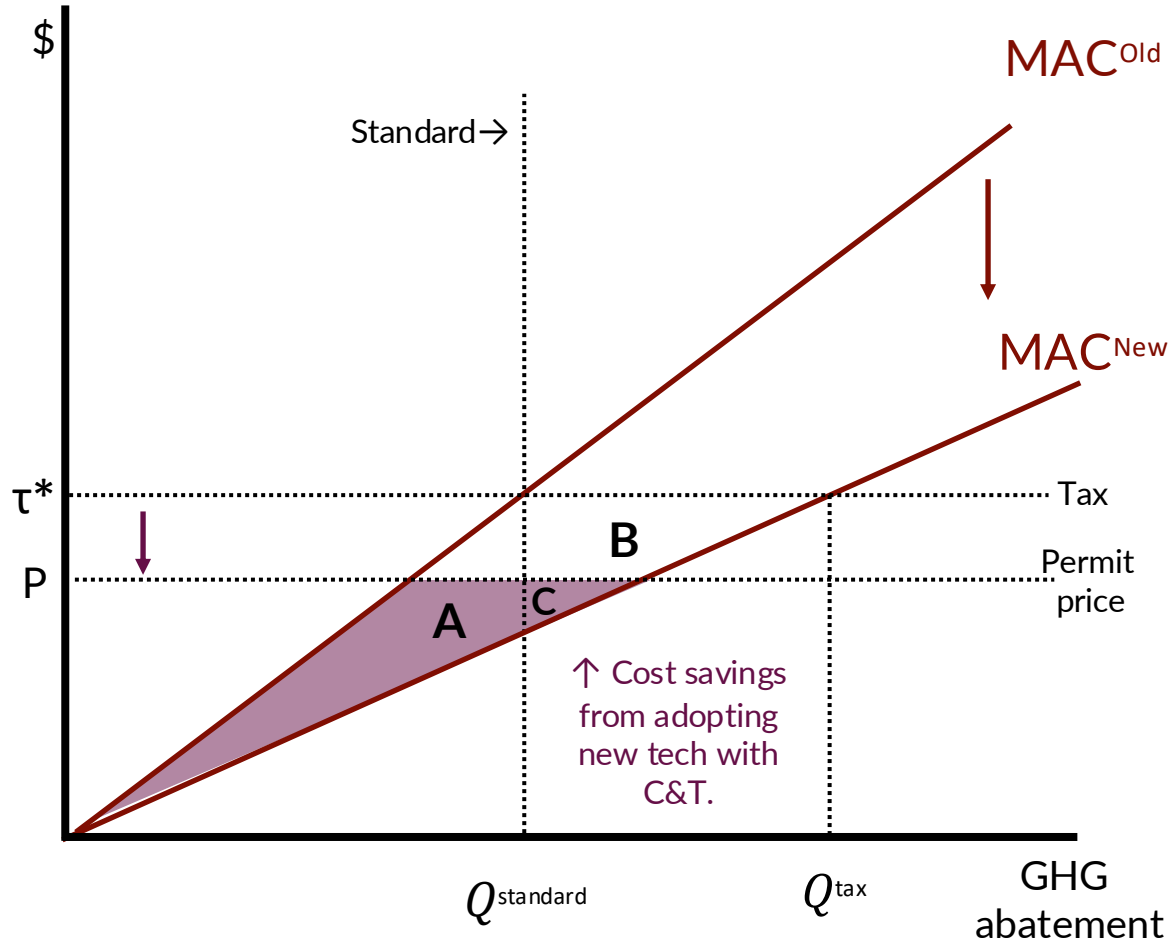
Innovation with taxes

- Imagine new technology becomes available that allows firms to generate same output with fewer CO_2 emissions.
- MAC curve pivots downwards.
- Under a tax:
  - Gain to the firm from adopting new technology is given by cost-savings represented by difference between MAC curves.
  - Area "B" represents additional incentive to adopt new tech./innovate.
  - More abatement occurs under tax.



Innovation with cap-and-trade

- Imagine new technology becomes available that allows firms to generate same output with fewer CO_2 emissions.
- MAC curve pivots downwards.
- Under cap-and-trade:
  - More difficult to illustrate because, in general, MACs are different for each firm…
  - Incentive to adopt still exists.
  - As more firms adopt, permit prices might decline (because it's cheaper to abate!)
  - Lower permit prices reduce incentive to innovate



Recap on policy instruments…

	Efficient? (i.e., maximizes net surplus)	Cost-effective? (i.e., minimizes costs)	Incentivizes innovation? (a.k.a. "dynamic efficiency")	Uncertainty?	Generates governmental revenue?
Command-and-control	**No**	**No**	**No**	**Certain Q outcome.**	**No**
Pigouvian taxes	**Yes**	**Yes**	**Yes**	**Uncertainty in Q.**	**Yes**
Cap-and-trade	**Yes**	**Yes**	**Yes, but weaker than taxes**	**Uncertainty in P.**	**Maybe** (e.g., by auctioning of initial permits)

Recap on policy instruments…

	Generates governmental revenue?
Command-and-control	**No**
Pigouvian taxes	**Yes**
Cap-and-trade	**Maybe** (e.g., by auctioning of initial permits)

- Generally, taxes are unpopular.
- A carbon tax would increase the cost (to consumers) of electricity, gasoline, etc.

- But…
- Carbon taxes can be designed to be *revenue neutral* by:
  - Providing lump-sum rebates (i.e., giving back money to residents)
  - Reduce other distortionary taxes (e.g., income taxes)
  - Invest in green infrastructure

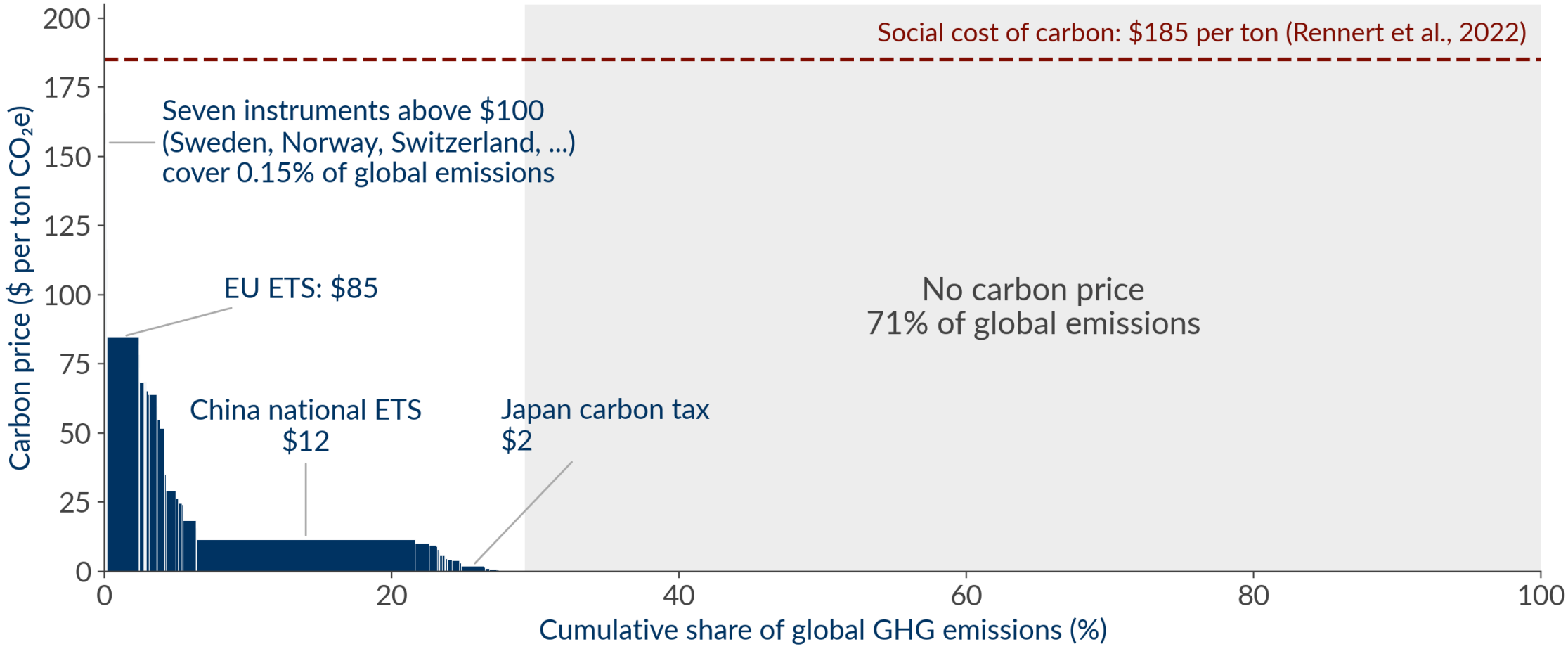
- **What you do with the revenue matters!**

Who prices carbon?

- 87 carbon taxes and trading systems are in operation around the world.
- Together, they cover **29%** of global GHG emissions.
- The average price is about **$21 per ton**.
  - Recall: the SCC is $185 per ton (Rennert et al., 2022).
- These programs raised $107 billion in government revenue in 2025.

Source: World Bank, State and Trends of Carbon Pricing 2026 and Carbon Pricing Dashboard.

Carbon prices around the world, 2026



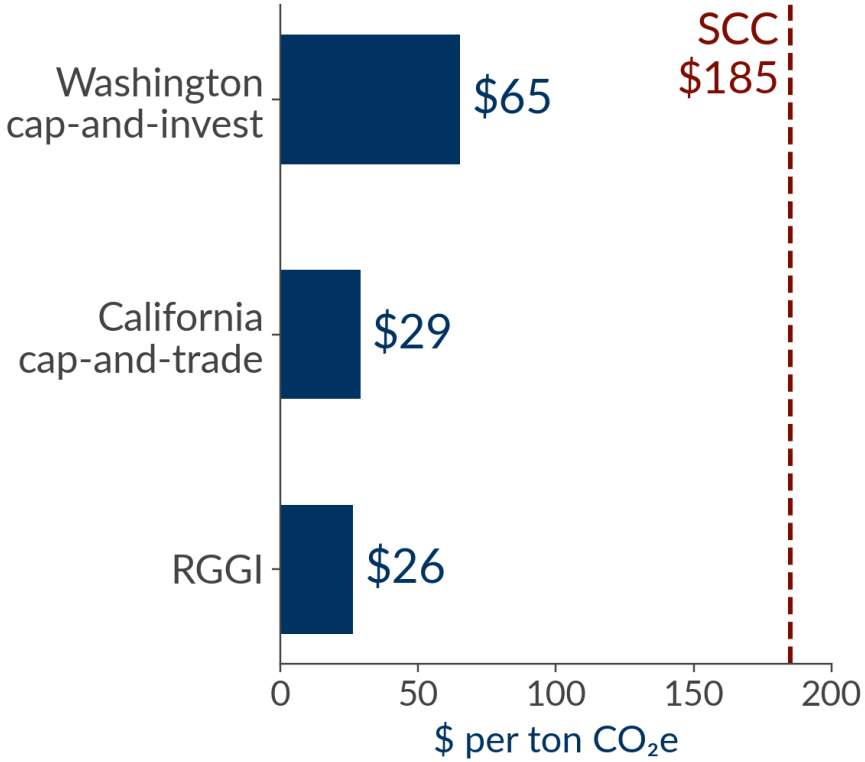
Each bar is one tax or trading system. Bar width is its share of global emissions. Prices as of April 1, 2026. Source: World Bank, State and Trends of Carbon Pricing 2026 and Carbon Pricing Dashboard.

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Carbon pricing in the US

- No federal carbon price.

- **California:** cap-and-trade since 2013.
  - Extended in 2025 through 2045.
- **Washington:** cap-and-invest since 2023.
  - Voters rejected a repeal in 2024.
- **RGGI:** power plants in northeastern states, since 2009.



Prices as of April 1, 2026. Source: World Bank Carbon Pricing Dashboard.

An aside on subsidies…

- The Inflation Reduction Act (IRA, 2022) relied mostly on **subsidies**.

- The 2025 tax law repealed most of them:
  - EV tax credits ended in September 2025.
  - Home efficiency and rooftop solar credits ended in December 2025.
  - Wind and solar credits end for projects in service after 2027.

- Subsidies can act like a (reverse) Pigouvian tax. But…
  - Subsidies require funding from somewhere else.
  - Subsidies “choose the winners” rather than letting prices decide.

Next class…

- Ghost Emissions - Round 7
- Review Keohane & Olmstead Chs. 8 and 9.
- Next class: We're going to play an in-class game about policy instruments.

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